

How Linguistic and Nonlinguistic Vocalizations Shape the Perception of Emotional Faces—An Electroencephalography Study

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Abstract

■ Vocal emotions are crucial in guiding visual attention toward emotionally significant environmental events, such as recognizing emotional faces. This study employed continuous EEG recordings to examine the impact of linguistic and nonlinguistic vocalizations on facial emotion processing. Participants completed a facial emotion discrimination task while viewing fearful, happy, and neutral faces. The behavioral and ERP results indicated that fearful nonlinguistic vocalizations accelerated the recognition of fearful faces and elicited a larger P1 amplitude, whereas happy linguistic vocalizations accelerated the recognition of happy faces and similarly induced a greater P1 amplitude. In

recognition of fearful faces, a greater N170 component was observed in the right hemisphere when the emotional category of the priming vocalization was consistent with the face stimulus. In contrast, this effect occurred in the left hemisphere while recognizing happy faces. Representational similarity analysis revealed that the temporoparietal regions automatically differentiate between linguistic and nonlinguistic vocalizations early in face processing. In conclusion, these findings enhance our understanding of the interplay between vocalization types and facial emotion recognition, highlighting the importance of cross-modal processing in emotional perception. ■

INTRODUCTION

Accurately recognizing others' emotions is crucial for establishing positive social relationships, and the integration of linguistic, prosodic, and visual emotional signals in a given social context enhances this recognition. Research has focused on elucidating the cerebral mechanisms of integrating linguistic, nonlinguistic, and visual emotional cues (Mauchand & Zhang, 2023; Tomasello, Grisoni, Boux, Sammler, & Pulvermüller, 2022). These include emphasizing how emotional signals from the voice and face are integrated (Nook, Lindquist, & Zaki, 2015; Pell et al., 2015). In nonemotional studies, Chen and Spence (2018) found that nonlinguistic vocalizations elicit cross-modal priming effects faster than spoken words. However, current research still lacks a comprehensive understanding of how these two types of vocalizations impact the recognition of emotional faces.

In general, communication sounds consist of linguistic and nonlinguistic elements. Nonlinguistic vocalizations are controlled by vocal cues like pitch, loudness, rhythm, and timbre, which reflect emotional states (Banse & Scherer, 1996). These vocalizations refer to sounds humans produce that do not contain linguistic information, such as laughing, crying, sighing, and screaming. Linguistic

vocalizations consist of the literal, symbolic meaning of the word (Kotz & Paulmann, 2007). Human linguistic vocalizations, including spoken words and sentences, carry linguistic information essential for communication. It should be noted that the term “linguistic vocalizations” does not include pseudo-utterances in this context (Pell et al., 2015). For this study, linguistic vocalizations are defined as those that contain explicit semantic information. Glaser and Glaser (1989) proposed that nonlinguistic vocalizations can access the semantic system directly, whereas linguistic vocalizations must pass through the lexical system before reaching the semantic system. Studies have suggested a potential fast route for perceiving emotional nonlinguistic vocalizations through the amygdala and possibly a partial overlap with the neural processes of emotional faces (Armony, Chochol, Fecteau, & Belin, 2007; Carroll & Young, 2005). However, no research has demonstrated a similar advantage for emotional and linguistic vocalizations (Liebenthal, Silbersweig, & Stern, 2016). In addition, there is a lateralization of activation for nonlinguistic vocalizations and linguistic vocalization: The left hemisphere is more prominent when processing linguistic information, whereas the right hemisphere is more strongly recruited when decoding prosodic information (George et al., 1996).

A few studies have focused on the interactions between facial expressions and emotional vocalizations. Pell,

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Jaywant, Monetta, and Kotz (2011) investigated the impact of prosody and semantics on facial recognition, confirming that sounds influence face discrimination in an emotion-congruent manner. Pell and colleagues (2015) compared the temporal dynamics of neural responses evoked by emotional speech prosody (pseudo-utterances emotional prosody) and nonlinguistic vocalizations (e.g., laughter) on face processing (judge the “facial expression represents an emotion”). They suggested that in vocal communication, nonlinguistic vocalizations are prioritized over speech-embedded emotions (larger N1 and P2 components) because of their unique evolutionary and neurofunctional origins. The multimodal affective information processing model proposed by Brück, Kreifelts, and Wildgruber (2011) suggested that integrating linguistic information with facial stimuli occurs in the left posterior superior temporal cortex, whereas nonlinguistic information processing occurs in the right posterior superior temporal cortex. In emotional research, there are three well-known hypotheses regarding the hemispheric lateralization of emotion: the right hemisphere hypothesis, which proposes that emotion and associated behaviors are predominantly lateralized to the right hemisphere (Demaree, Everhart, Youngstrom, & Harrison, 2005); the valence hypothesis, which proposes that the right hemisphere regulates negative emotion and related behaviors, whereas positive emotion and related behaviors are regulated by the left hemisphere (Killgore & Yurgelun-Todd, 2007). Recently, the emotion type hypothesis has suggested that the right hemisphere regulates primary emotions (related to self-preservation and reactions such as withdrawal, flight, or fight) and their related display behaviors. Meanwhile, social emotions (positive or negative in valence and associated with culturally dependent “display rules”) and their related display behaviors are regulated by the left hemisphere (Ross, 2021). Although numerous theories have explored the relationship between emotional sounds and faces, some aspects remain unclear. No studies directly compare the impact of linguistic and nonlinguistic vocalizations on emotional face recognition. Pell and colleagues (2015) compared the differences between nonlinguistic vocalizations and speech prosody, but the speech prosody lacked linguistic cues about emotion. However, they asked the participants to judge whether the faces contained any emotions, which made it difficult to directly examine how the two types of vocalizations affect the processing of faces with different emotions.

ERP studies indicated that emotional context affects multiple components of face processing (Luo, Feng, He, Wang, & Luo, 2010; Righart & De Gelder, 2006; Eimer & Holmes, 2002). Several cross-modality studies have demonstrated that emotional sounds enhance P1, regardless of whether people focus on geometric shapes (Brosch, Grandjean, Sander, & Scherer, 2009) or emotional pictures (Gerdes et al., 2013). The N170, a face-sensitive component, is assumed to reflect the stage of face structure processing. Natural emotional scenes (Hietanen & Astikainen,

2013) and emotional sentences (Li, Zhu, Ding, Ren, & Luo, 2019) modulate N170, as evidenced by the emotional context eliciting a larger N170. The modulation effects of emotional sounds on the face-sensitive N170 have not been reported (Liu et al., 2012). Early posterior negativity (EPN) reflects early selective attention and priority processing of salient information (Bublitzky, Gerdes, White, Riemer, & Alpers, 2014). Contextual factors like self-referential cues influence EPN (McCrackin & Itier, 2018). Late positive potential (LPP) predominantly reflects the congruency effect between context and facial emotion, enhanced when congruent emotions are present (Spreckelmeyer, Kutas, Urbach, Altenmüller, & Münte, 2006).

In addition to ERP analysis, representational similarity analysis (RSA) calculates the similarity between different representations in the brain under different stimuli or experimental conditions (Kriegeskorte, Mur, & Bandettini, 2008). Recently, RSA techniques have become widely used for ERP data. For instance, Li, Zhang, Liu, and Luo (2022) investigated the temporal course of the brain representation of emotional face recognition and found that arousal information is extracted earlier than valence. RSA aims to identify activation patterns that reliably distinguish between stimulus or condition attributes. If differences in neural activation patterns can be mapped to distinctions in stimulus attributes, then it can be inferred that these temporal or spatial neural patterns support distinctions in stimulus attributes. In the present study, we characterized the ERP data by constructing different representational dissimilarity matrix (RDM) from the stimulus attributes contained in the experimental conditions: vocalization type, face arousal, valence, and emotion congruence.

Affective priming is a widely used experimental paradigm for studying emotions (Gerdes et al., 2013). Affective priming can occur between face-to-face (Nook et al., 2015), word-to-face (Fugate, Gendron, Nakashima, & Barrett, 2018), and sound-to-face (Pell, 2005) stimuli. The priming cue produces a state of preparation for the target stimulus with the same emotional valence, thereby reducing RTs and increasing accuracy (ACC; Filippi et al., 2017; Schneider, Engel, & Debener, 2008; Averó & Calvo, 2006). This paradigm explored the association between linguistic or nonlinguistic vocalizations and different emotional faces. Combined with EEG techniques, it was possible to identify different electrophysiological signals for these associations. In the current study, we manipulated the priming vocalizations (linguistic and nonlinguistic vocalizations), faces (fearful, happy, and neutral), and congruence of emotions (congruent and incongruent) between the priming vocalization and target face. In this study, congruence refers to the alignment between the emotional information contained in the priming vocalizations and the facial emotion, specifically emotional congruence. We anticipated that vocalization would impact the early attentional components of facial perception. Therefore, vocalizations that attract individual attention more effectively under specific emotions elicit larger P1

amplitudes (Sauter & Eimer, 2010). The congruence of cross-modal emotional information affects the N170, EPN, and LPP components, with congruent conditions eliciting larger amplitudes (Lin & Liang, 2019). We anticipated identifying temporal and spatial differences in the representation of arousal and valence in RSA, hypothesizing that arousal representations would precede valence representations. We hypothesized that the RDM constructed from vocalization types would significantly correlate with the early EEG signals. Studies have shown that the differences between the two vocalizations can be distinguished during sensory processing and attention allocation (Pell et al., 2015). Finally, we hypothesized that the congruence RDM would not significantly affect the EEG signals, as participants were not instructed to assess the emotional congruence between the vocalizations and faces.

METHODS

Participants

The behavioral experiment recruited 32 healthy volunteers (13 male, aged 19–29 years; mean age: 23.28 years). The EEG experiment recruited 31 healthy volunteers (15 male, aged 18–28 years; mean age: 22.3 years). There were no overlapping participants between the behavioral experiment and the EEG experiment. On the basis of recent experiments that employed RSA and ERP methods to study emotional faces, it is appropriate to collect data from 31 participants in the EEG experiment (Żochowska & Nowicka, 2024; Li et al., 2022; Muukkonen & Salmela, 2022). A simulation conducted with the Superpower R package (Lakens & Caldwell, 2021) indicated that a sample size of 30 participants would be required to detect a significant interaction effect in a three-way 3 (Emotional Face) $\times$ 2 (Priming Vocalization) $\times$ 2 (Congruence) repeated-measures ANOVA, assuming a moderate effect size, with a power of 0.8 and an α level of .05. Simulation-based power analysis estimates the likelihood of detecting hypothetical effects in an ANOVA design by generating simulated data from factorial designs with defined parameters. All participants reported normal auditory and normal or corrected-to-normal visual acuity and were free of neurological or psychiatric problems. The participants in the behavioral experiment were remunerated \$5, and those in the EEG experiment were remunerated \$20. The study was approved by the Ethics Committee of Liaoning Normal University (China), and the latest version of the Declaration of Helsinki was used to conduct the investigation.

Stimuli

Face Stimuli

Sixty face pictures (20 faces each for fearful, happy, and neutral) were selected from the Chinese Facial Affective Picture System (Gong, Huang, Wang, & Luo, 2011), half male and half female. This study's face stimuli evaluation

data were sourced from the database. A one-way ANOVA was conducted on these pictures and showed that the three types of face pictures differed significantly in emotional valence, $F(2, 57) = 716.806, p < .001, \eta_p^2 = .962$. Compared with neutral ($M = 5.04, SE = 0.78$) and fearful ($M = 7.11, SE = 0.78$) faces, happy faces ($M = 2.92, SE = 0.78$) were rated more pleasant, whereas fearful faces were rated less pleasant ($ps < .001$). These faces significantly differed in their arousal levels, $F(2, 57) = 144.206, p < .001, \eta_p^2 = .835$ (neutral: 2.96 ± 0.16 , fearful: 6.45 ± 0.16 , happiness: 5.91 ± 0.16). Fearful and happy faces had higher arousal compared with neutral faces ($ps < .001$), but the difference between happy and fearful faces was not significant ($p = .051$). These images' size, background, spatial frequency, contrast, and luminance were consistent. Each face stimulus subtended a visual angle of $6.5^\circ \times 8.2^\circ$ (width $\times$ height).

Vocalization Stimuli

Nonlinguistic vocalization stimuli. Thirty emotional vocalizations, including 10 positive (e.g., laughing), 10 neutral (e.g., calmly ah ...), and 10 negative (e.g., screaming) vocalizations, were selected from the Montreal Affective Voices database (Belin, Fillion-Bilodeau, & Gosselin, 2008). There was no significant difference in the average vocalization duration ($M = 719.11, SE = 0.17$ msec), and the vocalization intensity was 60 dB.

Linguistic vocalization stimuli. Thirty two-character Chinese adjectives with positive (e.g., loyal), negative (e.g., fierce), and neutral (e.g., abstract) semantics were selected from the 12th Xinhua Dictionary for the experiments. The 30 words were then read by two male speakers in a neutral tone at a natural speed, resulting in 60 audio recordings. The vocalization materials were recorded in a quiet environment using high-quality microphones and Adobe Audition CS6 software, with a sampling rate of 44100 Hz and a transfer rate of 16 bytes per second. The participants recruited were Mandarin Chinese speakers and were paid for their work. The loudness of all vocalization materials was standardized to 60 dB using Praat software. There was no significant difference in the average vocalization duration for the three emotions, $F(2, 57) = 1.349, p = .268, \eta_p^2 = .045$ ($M = 816.56, SE = 14.28$ msec).

Arousal and Valence Measurements for Two Vocalizations

Seventeen healthy volunteers (6 male, mean age: 22.64 years) were recruited to assess the valence and arousal of two types of vocalizations using a 7-point scale (arousal: 1 = *not arousing*, 9 = *very arousing*; valence: 1 = *very unpleasant*, 9 = *very pleasant*; more detail in the Appendix). A two-way repeated-measures ANOVA on valence and arousal was conducted with Vocalization Category

(linguistic and nonlinguistic) and Emotion (negative, positive, and neutral) as within-subject factors. For valence, the results revealed a significant effect of Emotion, $F(2, 32) = 284.42, p < .001, \eta_p^2 = .947$. The negative (2.03 ± 0.11), positive (6.00 ± 0.13), and neutral vocalizations (3.88 ± 0.08) differed significantly from each other ($ps < .001$). The Vocalization Category's main effect was insignificant, with no difference between linguistic and nonlinguistic vocalizations, $F(1, 16) = 0.008, p = .929, \eta_p^2 = .001$. For arousal, a significant main effect of Emotion was found, $F(2, 32) = 77.94, p < .001, \eta_p^2 = .830$, with significantly higher ratings for positive (5.44 ± 0.22) and negative vocalizations (5.43 ± 0.15) than for neutral vocalizations ($3.08 \pm 0.29, ps < .001$). The Vocalization Category's main effect was insignificant, $F(1, 16) = 3.893, p = .066, \eta_p^2 = .196$.

Procedure

The experiment was conducted in a soundproof, softly lighted room. Participants were seated in a comfortable chair with their eyes 60 cm directly away from a 23-in. computer display. The screen resolution was 1920×1080 pixels, and the refresh rate was 60 Hz. Stimulus presentation was controlled, and behavioral responses were recorded using E-Prime 2.0. The behavioral and EEG experiment employed identical condition designs, each encompassing 12 conditions: 3 Emotional Faces (fearful, happy, and neutral) $\times$ 2 Priming Vocalizations (linguistic and nonlinguistic) $\times$ 2 Congruences (congruent and incongruent). In the incongruent condition, the priming vocalization used two other emotions (e.g., the emotion type of priming vocalizations under fearful-incongruent

conditions were happy and neutral), but the same-type vocalization as incongruent stimuli (details in the Appendix). In this study, congruence pertains to the congruence within the emotional dimension, and more precisely, it refers to the congruence between the emotional information encompassed in linguistic vocalizations (semantic information) or nonlinguistic vocalizations (emotional prosody) and the emotions of faces. To obtain reliable behavioral data, we required participants to respond quickly and accurately as soon as they saw the faces. Considering that keypressing creates additional noise in the EEG signal (Anwar et al., 2016), the response screen was delayed during the EEG experiment. We conducted the behavioral and EEG experiments separately, and the overall structure of both experiments was similar, with certain specific procedural differences.

In the behavioral experiment, each trial began with a random presentation of a white fixation cross (500–700 msec), followed by a 700- to 800-msec (depending on the length of the sound) sound presentation through headphones. A black screen appeared for 200 msec. Finally, a face image was displayed for 1000 msec, and the participant's task was to judge the emotional type of the face using the Space, J, and K keys on the keyboard, with response buttons counterbalanced across participants (Figure 1). Thirty-six trials were presented in the practice phase. The main experimental phase consisted of three blocks with ample breaks, each block containing 120 trials and each condition containing 30 trials, for 360 trials, with each condition presented randomly. The instructions also emphasized speed and ACC.

The procedures for EEG experiments and behavioral experiments were similar. The critical distinction is that

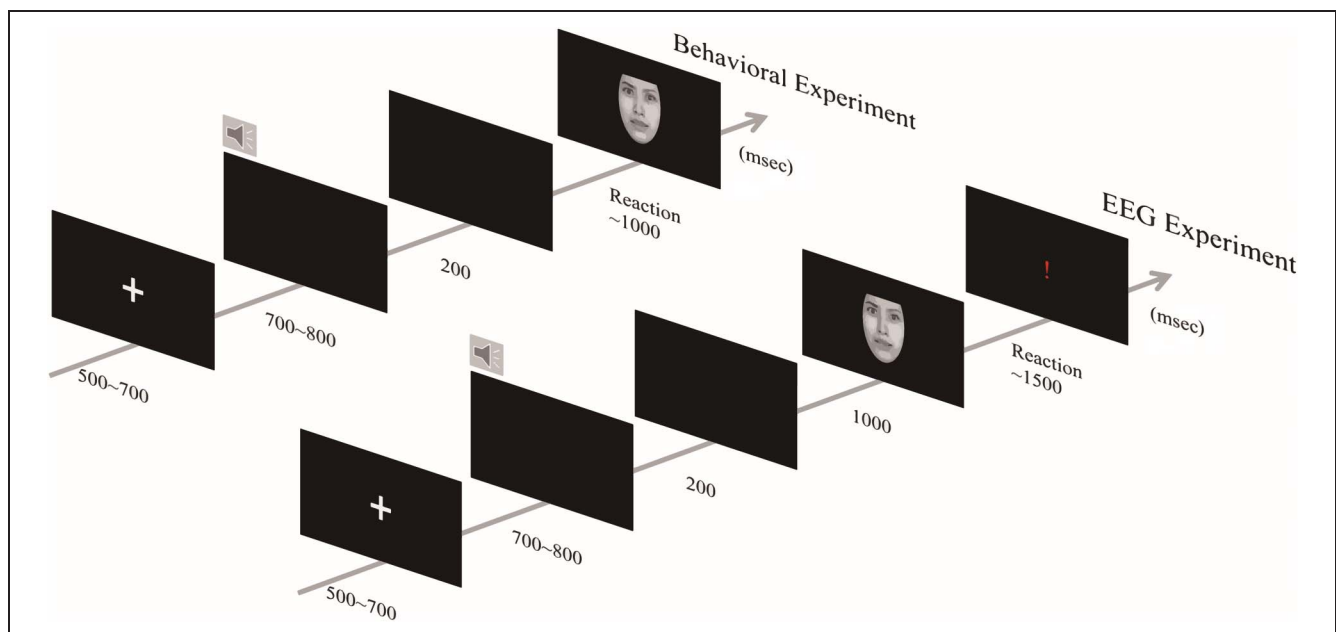


Figure 1. The figure shows the experimental procedure of a single trial in the behavioral experiment (upper) and EEG experiment (lower). In the behavioral experiment, participants could respond by pressing the key when they saw a face, whereas in the EEG experiment, participants could not respond until they saw a red exclamation mark on the screen after the face disappeared.

participants were required to respond only after 1500 msec following the presentation of a face, triggered by the appearance of a red exclamation mark on the screen (Figure 1). The EEG experiment consisted of three blocks with ample breaks in the middle, each block containing 240 trials, and each condition containing 60 trials, for 720 trials. Each condition was randomly presented to minimize habituation effects.

Behavioral Data Analysis

We first excluded RTs and ACC that exceeded 3 SDs. Then, we performed a three-way repeated-measures ANOVA with three Emotional Faces (fearful, happy, and neutral) $\times$ 2 Priming Vocalizations (linguistic and nonlinguistic) $\times$ 2 Congruence (congruent and incongruent) as within-subject factors by using SPSS 20.0. Degrees of freedom and p values were adjusted according to the Greenhouse-Geisser epsilon correction, and the Bonferroni correction ($p < .05$) was used for post hoc comparisons. The partial eta squared (η_p^2) was reported as a measure of effect size. No participants were excluded from the analysis.

EEG Recordings and Data Analysis

Preprocessing. EEG data were recorded using a 64-channel EEG recording system from Brain Products ERP, with tin electrodes placed on 64 scalp sites according to the extended 10–20 system and mounted on an elastic cap. EEG signals were recorded with a 0.01- to 100-Hz bandpass filter and a continuous sampling rate at 500 Hz/channel. The online reference was left mastoid, and the impedances of all electrode sites were kept below 5 k Ω . ERP data were preprocessed using EEGLAB (Delorme & Makeig, 2004). The raw data were first high-pass filtered at 0.1 Hz, low-pass filtered at 30 Hz, and off-line rereferenced to average. The data with large artifacts (exceeding $\pm 50 \mu\text{V}$) were removed. Components contaminated by eye blinks were detected and removed using the independent component analysis and ICLABEL algorithm (Pion-Tonachini, Kreutz-Delgado, & Makeig, 2019). The data were segmented into 1000-msec epochs time-locked to facial stimuli onset, starting 200 msec before stimuli onset. The epochs data were baseline corrected using the 200 msec before the stimuli onset. On average, about $12.8\% \pm 0.93\%$ of the trials per condition were rejected due to artifacts, and the rejection rate did not differ across conditions, $F(11, 360) = 0.028$, $p = 1.00$, $\eta_p^2 = .0008$. No participants were excluded from the EEG analysis.

ERP Analysis

Group-level ERP waveforms were computed by averaged single-participant ERP waveforms, and group-level scalp topographies at corresponding peak latencies were obtained by spline interpolation. These mean amplitudes

were used as the dependent variable in subsequent repeated-measures ANOVAs. According to the topographical distribution of the grand-averaged ERPs and the literature (Mück et al., 2020; Hinojosa, Mercado, & Carretié, 2015; Thom et al., 2014), the dominant ERP components were as follows: P1 (O1/O2/PO3/PO4/PO7/PO8) within the latency interval of 126–156 msec, N170 (PO7/PO8/P7/P8/P5/P6) within the latency interval of 176–226 msec, EPN (PO8/PO7/P7/P8/O1/O2) within the latency interval of 265–315 msec, and LPP (P1/P2/PZ/CP1/CP2/CPZ) within the latency interval of 350–650 msec. P1, N170, and EPN were analyzed using a four-way repeated-measures ANOVA with 3 Emotional Faces $\times$ 2 Priming Vocalizations $\times$ 2 Congruences $\times$ 2 Hemispheres (left and right). LPP were analyzed using a three-way repeated-measures ANOVA with 3 Emotional Faces $\times$ 2 Priming Vocalizations $\times$ 2 Congruences.

Decoding Analysis and RSA

Multivariate time-series decoding analysis was applied to the EEG data using support vector machine (SVM) classification and functions from the CoSMoMVPa toolbox (Oosterhof, Connolly, & Haxby, 2016; Chang & Lin, 2011) to explore further how the brain encodes this sensory information. We downsampled the segmented data to 250 Hz and subsequently performed binary classification to construct a neural RDM, where the pairwise decoding ACC was indicated as a measurement of the representative dissimilarity. SVM classifiers were trained and tested at each time point across all the electrodes. We randomly assigned each trial of one condition (12 experimental conditions) to one of the 10 splits and averaged the trials in each split to improve the signal-to-noise ratio. Training and testing were implemented using the leave-one-out method to avoid double-dipping: The classifier was trained on all but one averaged trial of one condition and tested on the left-one averaged trial. In this procedure, every averaged trial was excluded once, and the ACC was averaged across these repetitions. We performed the binary classification on all possible pairwise comparisons between conditions (66 pairs, calculated as $(12-1) \times 12/2$) to obtain decoding accuracies. All steps were repeated 100 times, and the accuracies over repetitions were averaged to construct a 12×12 neural RDM for each participant at each time point (300 neural RDMs per participant). In addition, we constructed arousal and valence RDM from the material measurement data of faces by calculating the Euclidean distance between faces to indicate their dissimilarity. We then constructed vocalization RDM according to the vocalization type, with the dissimilarity as 0 if it was the same type of vocalization and 1 otherwise. We also constructed a congruence RDM, with a dissimilarity of 0 if the emotions of the priming and target stimuli were congruent and 1 otherwise (Figure 2). We calculated the partial correlation between the different model RDMs and the neural RDMs at each time point for each

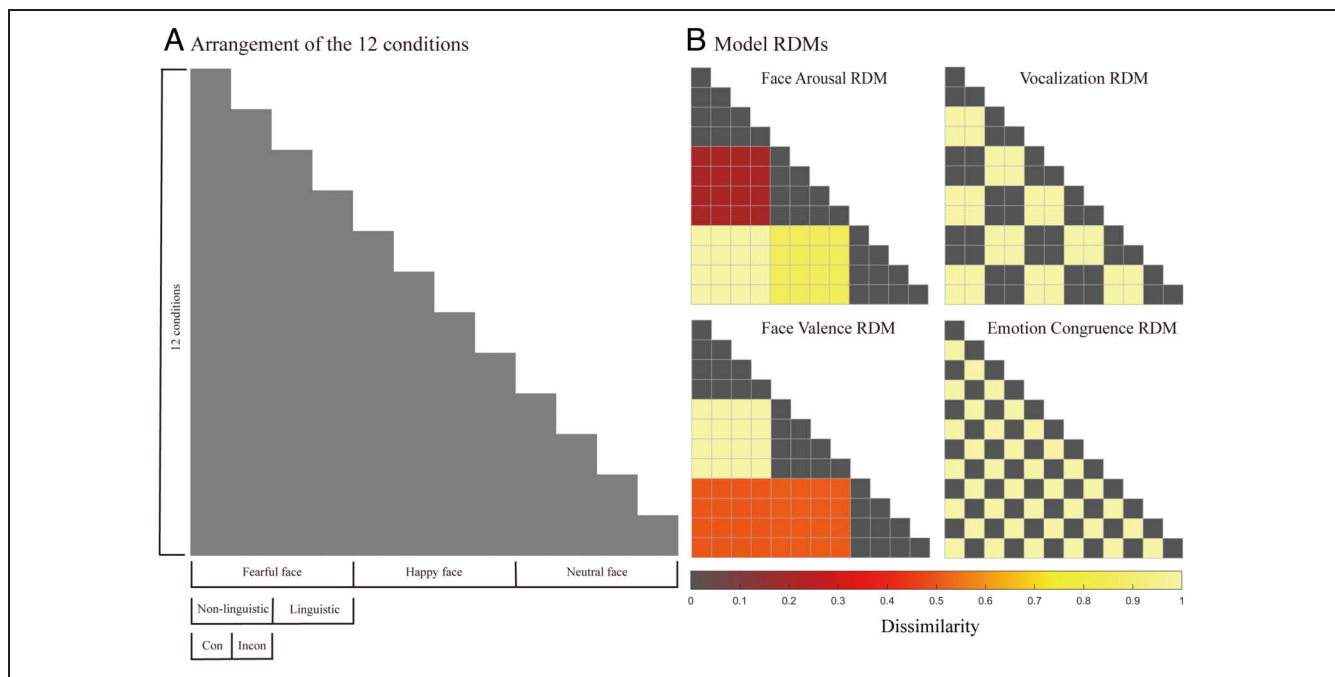


Figure 2. Schematic for model RDM. (A) To perform the RSA, we arranged the 12 conditions in this manner. (B) Model RDMs for different dimensions of stimuli. The colors represent the degree of similarity of a dimension between conditions. Zero means they are entirely similar, and 1 means they are not similar at all. Con = congruent; Incon = incongruent.

participant after excluding the other models. When calculating the partial correlations between the neural and valence RDM, the arousal RDM is excluded, which separates the contributions of these models from each other (Cichy & Pantazis, 2017; Giordano, McAdams, Zatorre, Kriegeskorte, & Belin, 2013). These partial correlation coefficients can be used as indicators of the time course of the different data dimensions.

Searchlight Analysis

We ran a multilevel regression searchlight RSA across the time points to investigate which specific electrodes contributed to the relationship between neural activity and the four RDMs. For each electrode, we constructed a neighborhood centered on the electrode and the four other electrodes around it. At each time point, the response pattern for each of the 12 conditions within the neighborhood was vectorized, and the neural phase anisotropy of each category pair was calculated using the Squared Euclidean distance to produce a 12×12 neural RDM. In other words, it was constructed based on the amplitude of each neighborhood rather than the decoding ACC. In addition, general linear regression was used to build the neural RDM from the four model RDMs for each neighborhood across time points. The regression coefficients of the four model RDMs were mapped to the central electrodes of the neighborhoods. The scalp distribution of the regression coefficients describes the whole-brain correspondence between the neural and model RDMs.

Temporal Generalization Analysis

We employed the temporal generalization approach of multivariate pattern analysis (King & Dehaene, 2014) to investigate whether different dimension representations were stable or changed over time. This approach trains a classifier on data at a specific time (training time) and tests it at all other time points (generalization time) to investigate if the representation remains stable. If the classifier can distinguish two conditions at the trained time and later time points, it indicates a stable representation format. Otherwise, it suggests a change in the format. We conducted temporal generalization analyses for each participant and three dimensions (all conditions, facial valence, and vocalization type) and then averaged the decoding accuracies of each participant. These resulted in three 300×300 matrices (−200 to 1000 msec to stimulus onset) that captured the classifier's generalization performance across time.

Statistical Inference

We performed permutation-based cluster size inference to examine the times the decoding ACC or correlation showed significant effects (Nichols & Holmes, 2002). The null hypothesis corresponded to a correlation coefficient of zero and a decoding ACC of 50%. Significant temporal clusters were specified as adjacent time points that exceeded a statistical boundary line (i.e., cluster-inducing threshold). The conditional labels of the data were permuted by random multiplication by +1 or −1 (i.e., the sign permutation test), and this process was repeated

5000 times to obtain the t value distribution of all possible permutations. On the basis of the t value distribution, the boundary line was determined using a sign permutation test. At each time point, the 95th percentile of the t value distribution was used as the clustering induction threshold (equivalent to $p < .05$, one-sided), and the 95th percentile of the maximum cluster size among all permutations was used as the significant cluster. This approach provided us with significant periods for decoding ACC and correlation. In the searchlight analysis, we first located the significant time points of each channel and then marked the significant channels on the topographic map at each time point.

Determination of Confidence Intervals (CIs)

Bootstrap tests estimated the statistical differences and 95% CIs for the onset, least, peak, and duration latencies between dimensions. The correlation coefficient was bootstrapped 1000 times, resulting in an empirical distribution of the onset (the minimum significant time point), peak (the significant time point at the maximum value), least (the least significant time point), and duration latencies (the duration of significant time). We defined the 95% CIs for these latencies as the 2.5th and 97.5th percentiles of the distribution. Latency differences were then calculated for any two dimensions in these 1000 bootstrap samples, resulting in an empirical distribution of latency differences. The p value was defined as the number of samples with differences greater or smaller than zero divided by the number of permutations (i.e., a two-sided

test) and corrected for multiple comparisons with a false discovery rate of 0.05.

RESULTS

All descriptive results for ERP data are expressed as mean $\pm$ SE.

Behavioral Results

ACC

The main effect of the Emotional Face was significant, $F(2, 30) = 4.165$, $p = .02$, $\eta_p^2 = .118$, with significantly higher ACC for the happy face relative to the fearful face ($p = .001$), and no difference between neutral and fearful faces. The main effect of Congruence was significant, $F(1, 31) = 11.551$, $p = .002$, $\eta_p^2 = .271$, with higher ACC for congruent conditions (Figure 3A).

RT

The main effect of the Emotional Face was significant, $F(2, 30) = 24.919$, $p < .001$, $\eta_p^2 = .446$, with significantly shorter RTs for happy faces than neutral and fearful faces and no difference between fearful and neutral faces. The main effect of Congruence was significant, $F(1, 31) = 12.373$, $p = .001$, $\eta_p^2 = .285$, with shorter RT under the congruent condition (Figure 3B).

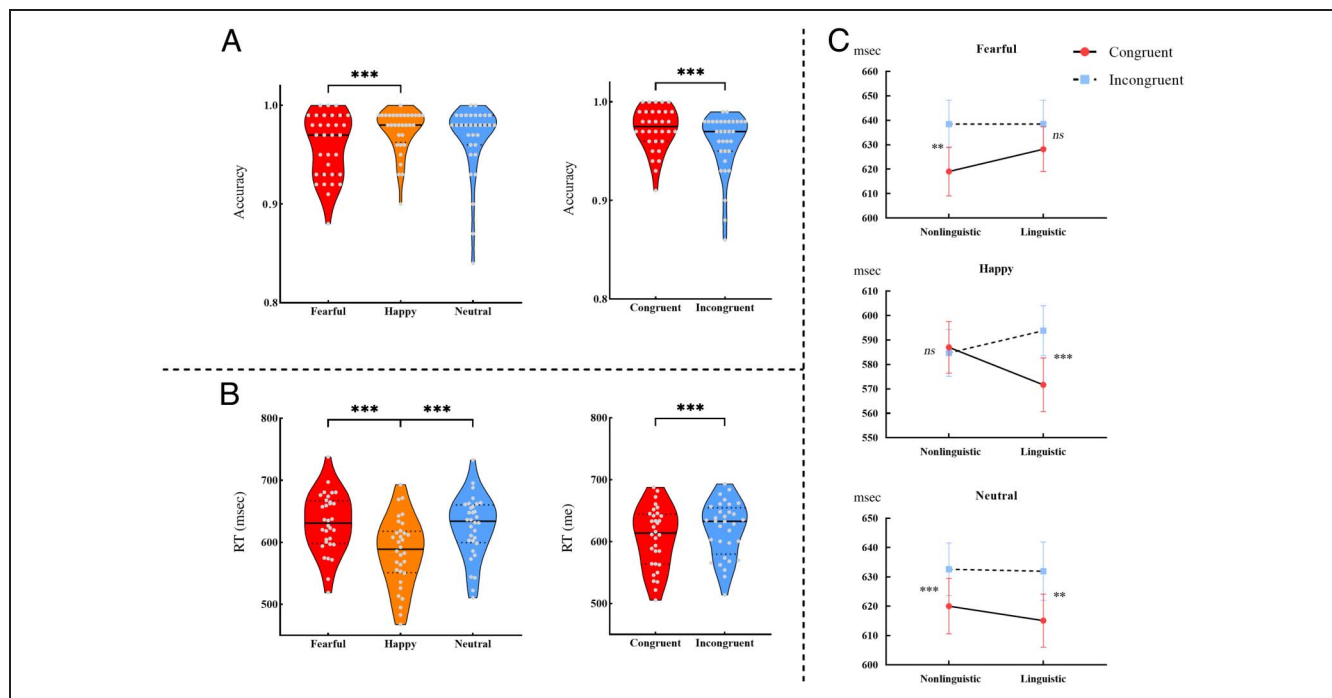


Figure 3. Behavioral experiment results. (A) Bar charts of ACC main effect of emotional face and congruence. (B) Bar charts of the RT main effect of Emotional Faces and Congruence. (C) Emotional Face $\times$ Priming Vocalization $\times$ Congruence three-factor interaction line chart of RT. The error bars represent SE. ns $p > .05$, * $p < .05$, ** $p < .01$, *** $p < .001$.

There was a significant three-way interaction effect between Emotional Face $\times$ Priming Vocalization $\times$ Congruence, $F(2, 30) = 6.604, p = .003, \eta_p^2 = .176$. Post hoc tests found that participants responded faster in the congruent condition under the fearful face when nonlinguistic vocalization was used as the priming stimulus. In the happy condition, participants responded faster in the congruent condition when linguistic vocalization was used as the priming stimulus. Participants responded faster in the congruent condition for the neutral face, regardless of the vocalization type (Figure 3C).

ERPs Results

P1

P1 results showed a significant main effect of Emotional Face, $F(2, 60) = 3.276, p = .045, \eta_p^2 = .098$, happy faces elicited significantly greater P1 wave amplitude than fearful faces ($p = .043$), and there was no difference between happy and neutral faces ($p = .561$, fearful: $3.319 \pm 0.430 \mu\text{V}$, happy: $3.539 \pm 0.412 \mu\text{V}$, and neutral: $3.435 \pm 0.397 \mu\text{V}$). The main effect of Congruence was significant, $F(1, 30) = 29.684, p < .001, \eta_p^2 = .497$; P1 induced by the incongruent condition was larger than the congruent condition (congruent: $3.144 \pm 0.407 \mu\text{V}$, incongruent: $3.718 \pm 0.420 \mu\text{V}$). The three-way interaction effect of Emotional Face $\times$ Priming Vocalization $\times$ Congruence was significant, $F(2, 60) = 5.568, p = .006, \eta_p^2 = .157$. Post hoc tests revealed that nonlinguistic vocalization elicited larger P1 than linguistic vocalization in the fearful face with the congruent condition ($p = .008$, nonlinguistic: $3.47 \pm$

$0.447 \mu\text{V}$, linguistic: $2.89 \pm 0.424 \mu\text{V}$). Linguistic vocalization elicited larger P1 amplitude than nonlinguistic vocalization in happy faces with congruent conditions ($p < .001$, nonlinguistic: $2.831 \pm 0.404 \mu\text{V}$, linguistic: $3.54 \pm 0.43 \mu\text{V}$). Under the neutral condition, there was no significant difference between linguistic and nonlinguistic vocalization, regardless of congruence (Figure 4A, Figure 5A).

N170

The significant main effect of Emotional Face was found in N170, $F(2, 60) = 9.464, p < .001, \eta_p^2 = .240$, with happy faces eliciting larger N170 amplitude than fearful and neutral faces, with no difference between fearful and neutral faces (fearful: $-2.958 \pm 0.454 \mu\text{V}$, happy: $-3.137 \pm 0.463 \mu\text{V}$ and neutral: $-2.804 \pm 0.451 \mu\text{V}$). The main effect of Congruence was significant, $F(1, 30) = 35.911, p < .001, \eta_p^2 = .545$, with larger N170 under the congruent condition (congruent: $-3.173 \pm 0.454 \mu\text{V}$, incongruent: $-2.759 \pm 0.456 \mu\text{V}$). The main effect of Hemisphere was significant, $F(1, 30) = 24.229, p < .001, \eta_p^2 = .447$, with a larger N170 amplitude in the right hemisphere (left: $-1.968 \pm 0.484 \mu\text{V}$ and right: $-3.965 \pm 0.510 \mu\text{V}$).

The two-way interaction effect of Emotional Face $\times$ Priming Vocalization was significant (Figure 4B), $F(2, 60) = 4.120, p = .021, \eta_p^2 = .121$. Under the fearful condition, the N170 elicited by linguistic vocalization was larger than nonlinguistic vocalization. In contrast, no difference was found between the two vocalizations in the happy and neutral conditions (fearful-nonlinguistic: $-2.829 \pm 0.469 \mu\text{V}$, fearful-linguistic: $-3.087 \pm 0.443 \mu\text{V}$, happy-nonlinguistic:

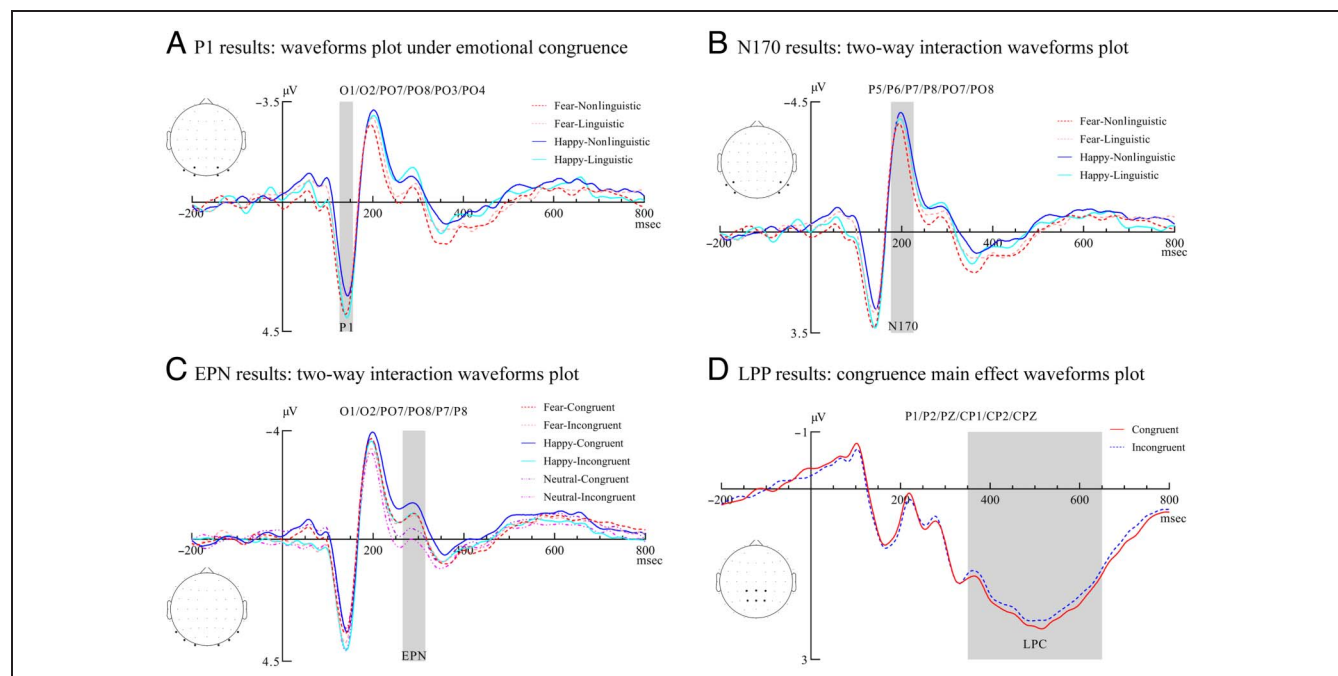


Figure 4. ERP waveforms of (A) P1, (B) N170, (C) EPN, and (D) LPP components. In the topographic map, the black dots represent the electrodes selected for this component, and the shaded area represents the selected time window.

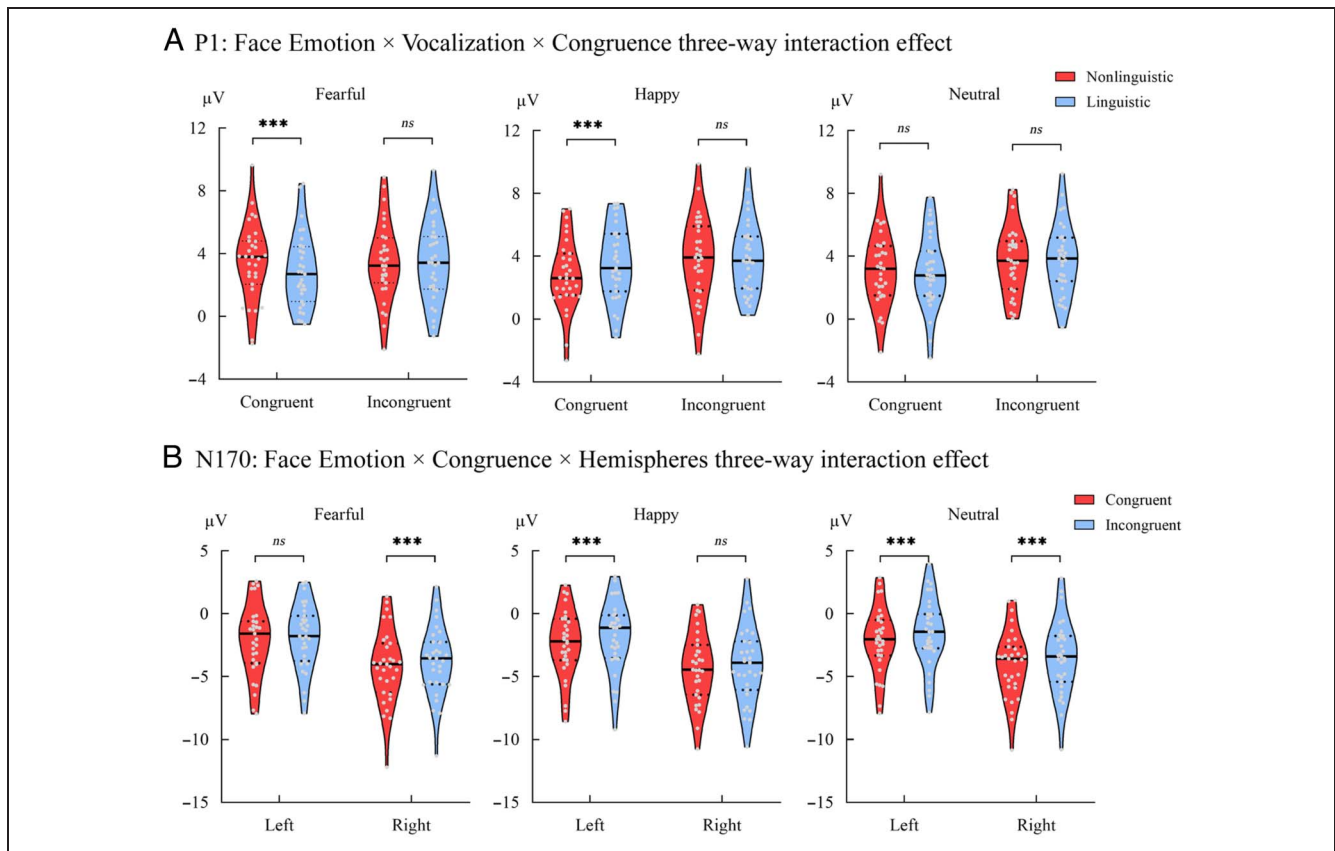


Figure 5. (A) P1 significant results of Emotional Face × Priming Vocalization Type × Congruence three-factor interaction. (B) N170 three-way interaction effect of Emotional Face × Congruence × Hemispheres. The error bars represent *SE*. *ns* $p > .05$, $*p < .05$, $**p < .01$, $***p < .001$.

-3.219 ± 0.469 μV , happy-linguistic: -3.055 ± 0.463 μV , neutral-nonlinguistic: -2.825 ± 0.449 μV , neutral-linguistic: -2.783 ± 0.46 μV).

As shown in Figure 5B, the three-way interaction effect of Emotional Face × Congruence × Hemisphere was significant, $F(2, 60) = 4.475$, $p = .015$, $\eta_p^2 = .130$; post hoc tests showed that in fearful condition, congruent condition elicited larger N170 in the right hemisphere ($p = .001$, congruent: -4.171 ± 0.532 μV , incongruent: -3.778 ± 0.504 μV), while in happy condition, congruent effect was found in the left hemisphere ($p = .001$, congruent: -2.385 ± 0.48 μV , incongruent: -1.842 ± 0.51 μV). In the neutral condition, a congruent effect was found in both hemispheres ($ps < .001$, left-congruent: -2.126 ± 0.473 μV , left-incongruent: -1.571 ± 0.501 μV , right-congruent: -4.035 ± 0.489 μV , right-incongruent: -3.483 ± 0.526 μV).

EPN

A significant main effect of Emotional Face was found in EPN, $F(2, 60) = 38.735$, $p < .001$, $\eta_p^2 = .564$, with happy faces larger than fearful faces and fearful faces larger than neutral faces (fearful: -0.728 ± 0.519 μV , happy: -0.964 ± 0.542 μV and neutral: -0.070 ± 0.530 μV). The main effect of Congruence was significant, $F(1, 30) = 10.194$, $p =$

$.003$, $\eta_p^2 = .254$, with congruent condition eliciting larger EPN than incongruent condition (congruent: -0.735 ± 0.526 μV and incongruent: -0.439 ± 0.532 μV). The two-way interaction effect of Emotional Face × Congruence was significant, $F(2, 60) = 3.424$, $p = .039$, $\eta_p^2 = .102$. The congruent conditions induced larger EPN amplitude in the happy and neutral faces and no difference in fearful faces (Figure 4C).

LPP

LPP results showed a significant main effect of Congruence, $F(1, 30) = 4.286$, $p = .047$, $\eta_p^2 = .125$, with congruent condition eliciting a larger LPP than incongruent condition (Figure 4D).

Decoding and RSA Results

To further elucidate the entire cognitive process, we trained and tested each pair of conditions at each time point across all channels using the SVM classifier for each participant. We averaged all values in the ACC matrix for each time point. The results showed that the 12 conditions could be significantly distinguished starting at 4 msec ($t(30) = 1.91$, Cohen's $d = 0.697$, cluster-corrected sign permutation test, cluster definition threshold $p < .05$,

cluster-corrected significance level $p < .05$) with an average ACC of 53.21%, peaking at 292 msec and decreasing at 760 msec. These results show how EEG responses are discriminated between the 12 conditions and provide an essential basis for the RSA.

The RSA results revealed the time course of the representations of arousal (Figure 6A, significant time points: 204–328 msec, 336–584 msec, 772–904 msec, and 916–968 msec), valence (92–976 msec), vocalization type (0–200 msec and 216–264 msec), and congruence

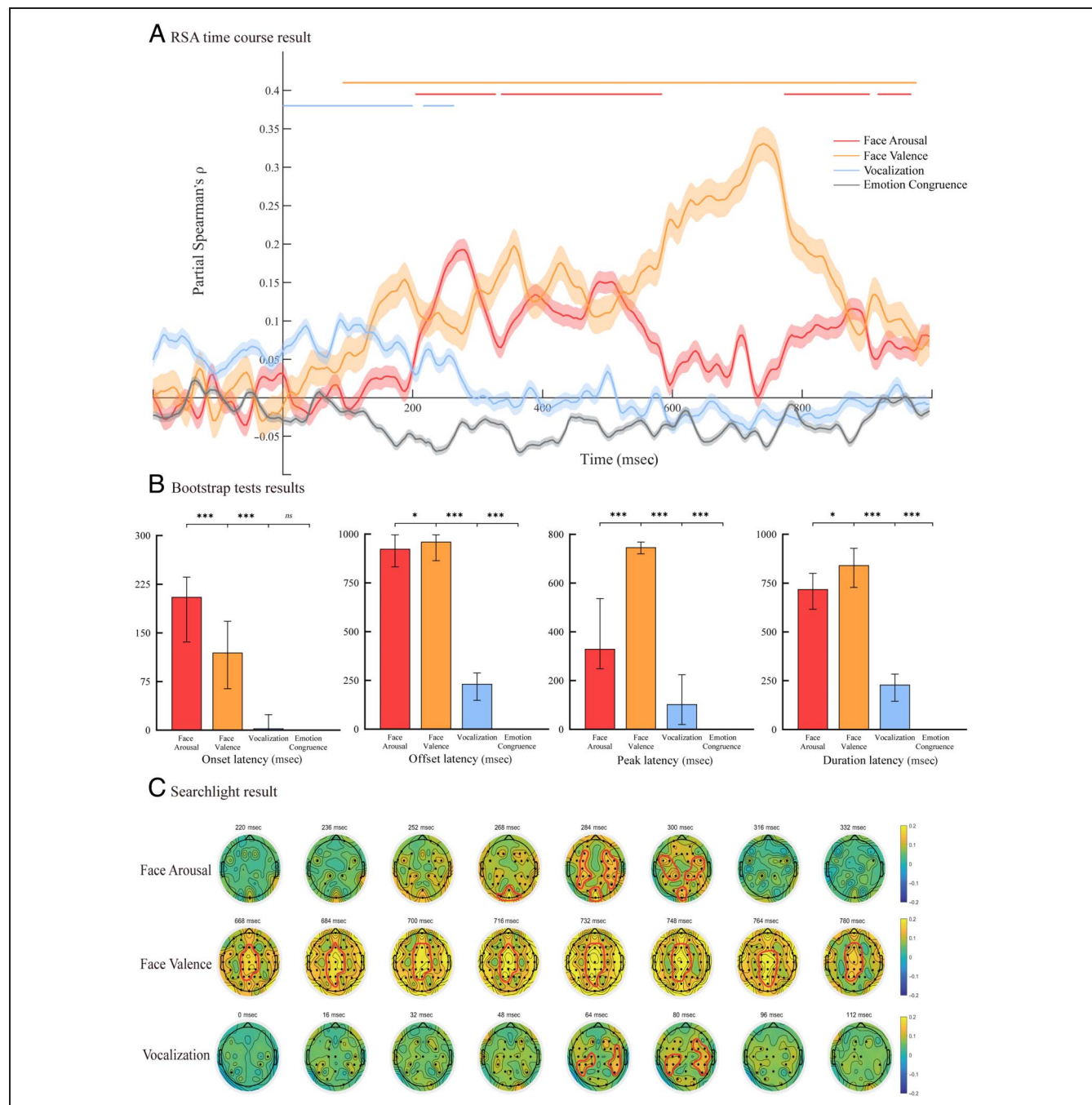


Figure 6. RSA results. (A) Time course of partial Spearman correlations between neural RDMs and model RDMs for arousal (red), valence (orange), vocalization (blue), and congruence (gray). The lines above the horizontal axis indicate significant time points using the cluster-corrected sign reciprocity test, and the line colors represent the respective dimensions (cluster definition threshold $p < .05$, cluster-corrected significance level $p < .05$). (B) Onset, least, peak, and duration latencies for four dimensions. Error bars indicate bootstrapped 95% CIs. ns $p > .05$, $*p < .05$, $**p < .01$, $***p < .001$ (two-sided bootstrap test, false discovery rate-corrected). (C) Plots display the beta values in GLM as topographic distribution. Arousal topographic display period: 220–332 msec, valence: 668–780 msec, vocalization: 0–112 msec; each interval is 16 msec. The black dots in the topographic maps represent channels reaching significance ($p < .05$, cluster-corrected sign permutation test). The red line highlights the main significant areas.

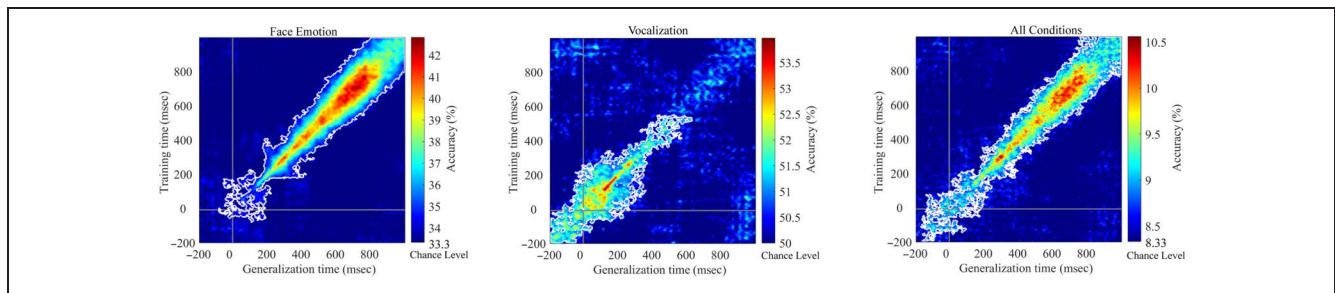


Figure 7. Temporal generalization result. Temporal cross-classification matrix for different dimensions. The y axis represents the classifier training time corresponding to stimulus onset. The x axis represents the classifier generalization time. The color scale indicates the cross-classification ACC for each training and generalization time combination. The white outline represents significant clusters ($p < .05$, cluster-corrected sign permutation test).

(no significant time point). The vocalization information was first extracted from the EEG signal, followed by extraction of the face valence information at approximately 119.6 msec (Figure 6B, $p < .001$). Face arousal information was extracted at 205.4 msec ($p < .001$). Compared with arousal and valence, the vocalization type appeared earlier but with a shorter duration ($p < .001$), implying that vocalization priming has an early and rapid impact on face processing. Arousal (about 329.7 msec) reached its peak significantly earlier than valence (about 746.7 msec, $p < .001$).

Searchlight Result

Figure 6C shows topographies at significant time points corresponding to each dimension's peak time points. The topography of the vocalization showed that the channels located in the temporal and temporoparietal regions were significant. In the arousal dimension, some of the channels in the occipital and temporoparietal regions were significant, and the vast majority of channels in the parietal and occipital regions were significant in the valence dimension. No significant channels were found in congruence dimensions.

Temporal Generalization Result

The temporal generalization of emotion dimension appeared at about 150–180 msec and lasted around 900 msec. The brain activity between 200 and 400 msec is relatively stable, as the neural representations at 200 msec (training set) can correctly distinguish the representations at 400 msec (testing set). The temporal generalization of vocalization dimension starts at about –50 to 0 msec and ends at around 350 to 400 msec, and temporal generalization is stable at 0 to 200 msec. We could distinguish the difference between the two types of vocalizations in brain representation patterns that persisted for up to 300 msec. The temporal generalization of the condition dimension began at around 200–400 msec and lasted until about 800 msec (Figure 7).

DISCUSSION

This study examined the temporal dynamics of priming effects from emotional linguistic and nonlinguistic vocalizations on facial expression recognition. Behaviorally, fearful nonlinguistic vocalizations facilitated faster recognition of fearful faces, whereas happy linguistic vocalizations enhanced the recognition of happy faces. Nonlinguistic vocalizations elicited a larger P1 component for fearful faces, whereas happy faces were associated with a larger P1 in response to linguistic vocalizations. In the N170 component, congruency effects were observed in the right hemisphere for fearful faces and in the left hemisphere for happy faces. The RSA revealed that information regarding vocalization type difference is retained in working memory during early face processing, with searchlight analyses indicating regional variations in representation. The representation patterns of arousal and valence support the circumplex model of affect, aligning with findings from studies involving the passive viewing of emotional faces.

The behavioral results indicate that emotional information conveyed through the auditory channel enhances the perception of emotional faces. Under emotionally congruent conditions, participants exhibited higher ACC and faster RTs in identifying emotional expressions. This effect was corroborated by ERP data, which showed that congruent conditions elicited larger amplitudes in the P1, N170, and EPN components, whereas incongruent conditions resulted in larger LPP amplitudes. The increased P1 amplitude emphasizes the role of auditory expressions in influencing visual perception and directing attention to faces (Gupta, Kujawa, & Vago, 2019) while also affecting early facial structural processing (N170; Pell et al., 2022). Arousing emotional stimuli are known to increase the EPN within the 150- to 350-msec latency range following the presentation of a face (Schupp & Kirmse, 2021), particularly for fearful and angry expressions (Schindler & Bublatzky, 2020). When the vocalization is congruent with the facial expression, it enhances the individual's emotional arousal, resulting in a stronger EPN. The larger LPP amplitude suggests challenges integrating facial details with previous events stored in working memory,

leading to sustained attention on the target and increased cognitive effort (Stolz, Endres, & Mueller, 2019; Diéguez-Risco, Aguado, Albert, & Hinojosa, 2015). The LPP findings reflect the need for greater cognitive resources when individuals attempt to reconcile conflicting emotional information, particularly in adapting to complex emotional contexts (Schupp, Flaisch, Stockburger, & Junghöfer, 2006). These results suggest that auditory emotional cues enhance early-stage processing of facial emotions in the visual system and improve perceptual efficiency through cross-modal emotional congruence.

The three-way interaction results for RT suggest that, under emotionally congruent conditions, nonlinguistic vocalizations accelerate the perception of fearful faces, whereas linguistic vocalizations enhance the speed of recognizing happy faces. P1 results further demonstrated that nonlinguistic vocalizations elicited larger P1 amplitudes when recognizing fearful faces, whereas linguistic vocalizations produced larger P1 amplitudes when recognizing happy faces. We propose that this vocalization–emotion preference stems from biases in early attentional allocation (Wang, Tang, Zhang, & Yang, 2022). This pairing of vocalizations with specific emotions reflects the brain’s early allocation of attentional resources, which facilitates more efficient recognition and response to emotional threats or rewards based on the type of vocalization.

A common perspective on the N170 function posits that facial sensitivity modulation reflects the early stages of facial configuration and structural processing (Gao, Conte, Richards, Xie, & Hanayik, 2019; Deffke et al., 2007). Previous priming studies have shown that visual and contextual environments influence the demands on facial structural processing (Diéguez-Risco et al., 2015; Hietanen & Astikainen, 2013; Righart & de Gelder, 2008). The three-way interaction results for the N170 revealed a consistency difference for fearful faces in the right hemisphere and happy faces in the left hemisphere. We propose that during the face configuration stage, the brain compares cross-modal information for different emotions within the left and right hemispheres, respectively (Hinojosa et al., 2015; Brück et al., 2011). On the basis of both behavioral and ERP results, we infer that fearful faces are more sensitive to nonverbal vocal priming during face recognition’s early attentional allocation phase. In contrast, happy faces are more sensitive to verbal sounds, capturing greater attentional resources. During the face configuration stage, the right hemisphere automatically engages in cross-modal comparisons for fearful faces, whereas the left hemisphere is more sensitive to consistency differences in happy faces. These findings are consistent with the emotion type hypothesis: The right hemisphere is primarily involved in processing survival-related emotions such as withdrawal, flight, or fighting, whereas the left hemisphere is associated with social emotions (Ross, 2021; Ross, Thompson, & Yenkosky, 1997).

RSA enables the modeling of complex mental representations, capturing the multidimensional characteristics of

information processing (Grootswagers, Kennedy, Most, & Carlson, 2020; Dobs, Isik, Pantazis, & Kanwisher, 2019). The RSA results for vocalization type indicated that, during the earliest stages of facial processing (0–232.5 msec), the brain represented the two types of vocalizations differently. Searchlight analysis revealed that this differentiation originated from electrodes in the temporoparietal region. Different types of vocalizations modulate attention to emotional faces differently during early processing stages. Specifically, the brain’s responses to vocalizations shape how individuals allocate attention to facial features, facilitating the precise integration of relevant auditory information and enhancing perceptual efficiency (Brück et al., 2011). Although participants were not explicitly required to focus on the auditory stimuli, the results demonstrate that the brain automatically distinguished between the two types of vocalizations. Temporal generalization results further supported this finding, showing stable decoding between the two types of vocalizations (training and generalization) within the 0- to 200-msec range. Previous research has demonstrated that distinctions between speech prosody (in pseudo-utterances) and nonlinguistic vocalizations can occur during the sensory and attention distribution phases (Pell et al., 2015). Our findings suggest that the brain processes visual stimuli and integrates auditory information in the early stages of facial processing. This multidimensional representational mechanism likely reflects the brain’s collaborative processing of multisensory information during perception.

The representation patterns of arousal and valence revealed that arousal primarily influences the early stages of emotional face processing, whereas valence is more involved in the later cognitive processing stages. These findings are consistent with previous studies that utilized passive viewing of emotional faces (Li et al., 2022). Popal, Wang, and Olson (2019) highlighted that effectively differentiating the neural activities associated with these two dimensions remains a challenge, as emotional processing involves multiple overlapping neural processes, making it difficult to isolate specific emotional dimensions temporally or spatially. Basic emotion theory proposes that distinct neural systems support each emotion category. However, an increasing number of studies support the circumplex model of affect, which posits that emotions are not discrete categories but are instead represented along the dimensions of valence and arousal, supported by distinct neurophysiological systems. Recent EEG studies have also suggested that emotion classification is not based on discrete categories (Li et al., 2022; Muukkonen, Ölander, Numminen, & Salmela, 2020). A growing body of fMRI research supports the circumplex model (Li, Li, Hu, Yang, & Luo, 2024; Bush, Privratsky, Gardner, Zielinski, & Kilts, 2018; Yuen et al., 2012). Recent findings indicate that arousal representation primarily occurs in the parahippocampal gyrus, whereas valence representation relies on the connection between primary visual cortex and the parahippocampal gyrus (Li et al., 2024). This perspective

challenges basic emotion theory, suggesting that dimensional representations of emotion provide a more comprehensive explanation for the diversity and flexibility of emotional experiences than single-category neural models (Calvo & Nummenmaa, 2016). In emotional face processing, the interaction between arousal and valence likely extends throughout the entire emotional processing sequence, influencing brain network activation at various stages. By integrating multimodal representational analysis with high temporal resolution, future research can further explore how these two emotional dimensions are encoded explicitly in the brain and more clearly distinguish their roles in different emotional tasks.

The findings of this study have advanced our understanding of the temporal dynamics involved in facial expression processing. However, several limitations must be acknowledged. First, we examined only two specific emotions—fear and happiness—whereas other emotional states, such as anger, sadness, and surprise, were not included. Future research should explore a broader spectrum of emotional expressions better to understand the interactions between vocalizations and facial emotions. Second, the linguistic vocalization stimuli in this study were recorded exclusively by male speakers, limiting our

ability to investigate potential gender effects. Previous research suggests that gender may significantly influence emotional perception. For instance, individuals often exhibit more robust empathetic responses to the crying sounds of the opposite sex (Chun, Park, Park, & Kim, 2012). The exclusion of female vocalizations in this study may have overlooked essential differences in cross-modal emotional processing.

Conclusion

In summary, this study investigated the effects of emotional linguistic and nonlinguistic vocalizations on facial expression recognition, utilizing multidimensional EEG decoding to uncover the precise temporal dynamics of contextual and facial information processing. Our findings revealed attention biases toward vocalization–emotion facial pairings and demonstrated that emotional representations in the brain align with the circumplex model of affect. This study provides novel neural evidence for the interaction mechanisms between emotional faces and vocalizations, further elucidating the spatiotemporal characteristics of cross-modal emotional integration and its impact on emotion perception.

APPENDIX

Material

Table A1. Arousal and Valence Ratings of Linguistic Vocalizations

	<i>Valence</i>	<i>Arousal</i>
<i>Positive Vocalizations</i>		
Brave	6.38	5.85
Confident	6.33	5.36
Delightful	6.35	5.27
Glad	6.29	5.56
Happy	6.27	5.47
Loyal	6.24	5.65
Optimistic	6.44	5.68
Outstanding	6.35	5.44
Pleasant	6.30	5.29
Successful	6.21	5.47
<i>Negative Vocalizations</i>		
Aggressive	1.59	5.03
Brutal	1.97	4.56
Cruel	1.38	5.74
Filthy	1.53	4.82
Defective	1.47	5.59
Inferior	1.53	5.21
Scoundrel	1.71	5.03
Shameless	1.41	5.77
Terrifying	1.86	5.03
Ugly	1.86	5.27
<i>Neutral Vocalizations</i>		
Aggressive	4.12	3.21
Brutal	3.77	3.00
Cruel	4.03	2.74
Filthy	4.00	2.56
Defective	3.97	3.21
Inferior	4.06	2.88
Scoundrel	3.88	2.36
Shameless	4.12	2.97
Terrifying	3.94	2.65
Ugly	3.97	2.74

Table A2. Arousal and Valence Ratings of Nonlinguistic Vocalizations

	<i>Valence</i>	<i>Arousal</i>
<i>Positive Nonlinguistic Vocalization</i>		
Clip 1	5.79	5.35
Clip 2	5.59	5.03
Clip 3	5.35	5.29
Clip 4	5.50	5.09
Clip 5	5.65	4.88
Clip 6	5.35	5.09
Clip 7	6.03	5.53
Clip 8	5.76	6.29
Clip 9	6.09	5.91
Clip 10	5.79	5.35
<i>Negative Nonlinguistic Vocalization</i>		
Clip 1	2.03	5.88
Clip 2	2.47	5.12
Clip 3	1.91	6.00
Clip 4	1.29	6.56
Clip 5	2.94	5.88
Clip 6	3.91	4.65
Clip 7	2.74	5.06
Clip 8	2.41	5.50
Clip 9	2.26	6.15
Clip 10	2.38	5.76
<i>Neutral Nonlinguistic Vocalization</i>		
Clip 1	3.76	3.12
Clip 2	3.91	3.12
Clip 3	3.88	3.26
Clip 4	3.71	3.32
Clip 5	3.94	3.56
Clip 6	3.71	3.24
Clip 7	3.76	3.24
Clip 8	3.68	3.26
Clip 9	3.88	3.71
Clip 10	3.53	3.56

Method

Regarding Stimulus Combinations under Incongruent Conditions

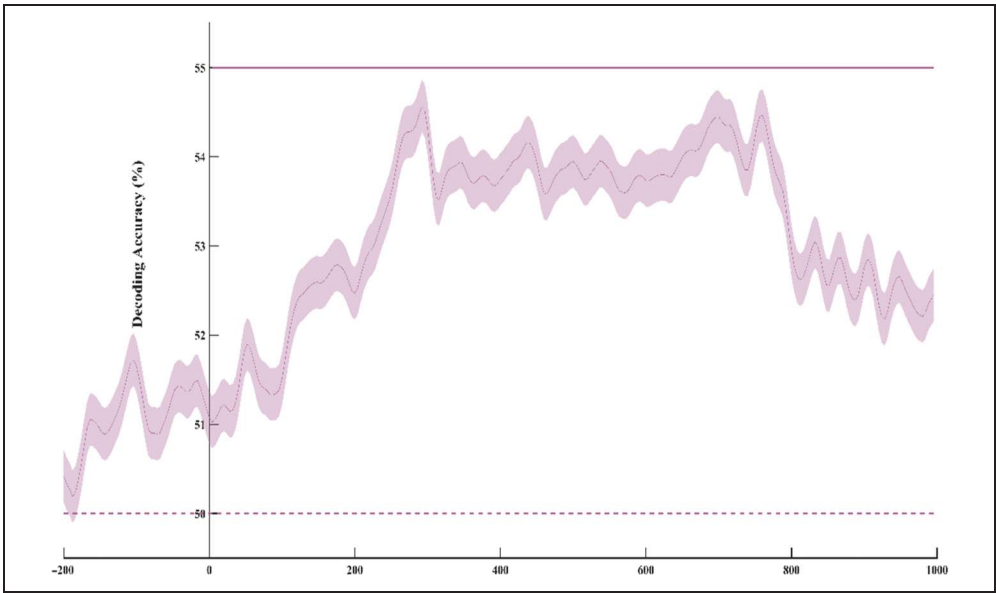
In the incongruent condition, the priming vocalization used two other emotional but the same vocalization type (e.g., linguistic vocalization corresponding to linguistic vocalization) vocalization as incongruent stimuli. For example, in the EEG study, there were 60 trials in the

fearful-face nonlinguistic vocalization incongruence condition, then each fearful face (20 pieces) was repeated three times, the priming vocalization consisted of nonlinguistic positive (10 pieces) and neutral (10 pieces) vocalization, and they were also repeated three times. Finally, their direct combination was completely randomized. The stimulus combinations for all conditions are shown in the table below.

Condition	Face Stimulus	Vocalization Stimulus
Fear-nonlinguistic-congruent	Fearful face	Negative nonlinguistic vocalization
Fear-nonlinguistic-incongruent	Fearful face	Positive & neutral nonlinguistic vocalization
Fear-linguistic-congruent	Fearful face	Negative linguistic vocalization
Fear-linguistic-incongruent	Fearful face	Positive & neutral linguistic vocalization
Happy-nonlinguistic-congruent	Happy face	Positive nonlinguistic vocalization
Happy-nonlinguistic-incongruent	Happy face	Negative & neutral nonlinguistic vocalization
Happy-linguistic-congruent	Happy face	Positive linguistic vocalization
Happy-linguistic-incongruent	Happy face	Negative & neutral linguistic vocalization
Neutral-nonlinguistic-congruent	Neutral face	Neutral nonlinguistic vocalization
Neutral-nonlinguistic-incongruent	Neutral face	Positive & negative nonlinguistic vocalization
Neutral-linguistic-congruent	Neutral face	Neutral linguistic vocalization
Neutral-linguistic-incongruent	Neutral face	Positive & negative linguistic vocalization

Result

Figure A1. Decoding ACC. The lines above the horizontal axis indicate significant time points using the cluster-corrected sign reciprocity test (cluster definition threshold $p < .05$, cluster-corrected significance level $p < .05$).



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Data Availability Statement

All data generated/analyzed during this study are available from the corresponding authors upon reasonable request.

Author Contributions

Junyu Liang: Conceptualization; Data curation; Formal analysis; Methodology; Visualization; Writing—Original draft; Writing—Review & editing. Mingming Zhang: Writing—Original draft; Writing—Review & editing. Lan Yang: Writing—Original draft; Writing—Review & editing. Yiwen Li: Writing—Review & editing. Yuchen Li: Writing—Review & editing. Li Wang: Writing—Review & editing. Hongying Li: Writing—Review & editing. Jun Chen: Supervision; Writing—Review & editing. Wenbo Luo: Funding acquisition; Supervision; Writing—Original draft; Writing—Review & editing.

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Diversity in Citation Practices

Retrospective analysis of the citations in every article published in this journal from 2010 to 2021 reveals a persistent pattern of gender imbalance: Although the proportions of authorship teams (categorized by estimated gender identification of first author/last author) publishing in the *Journal of Cognitive Neuroscience (JoCN)* during this period were $M(\text{an})/M = .407$, $W(\text{oman})/M = .32$, $M/W = .115$, and $W/W = .159$, the comparable proportions for the articles that these authorship teams cited were $M/M = .549$, $W/M = .257$, $M/W = .109$, and $W/W = .085$ (Postle and Fulvio, *JoCN*, 34:1, pp. 1–3). Consequently, *JoCN* encourages all authors to consider gender balance explicitly when selecting which articles to cite and gives them the opportunity to report their article's gender citation balance.

REFERENCES

- Anwar, A. R., Muthalib, M., Perrey, S., Galka, A., Granert, O., Wolff, S., et al. (2016). Effective connectivity of cortical sensorimotor networks during finger movement tasks: A simultaneous fNIRS, fMRI, EEG study. *Brain Topography*, 29, 645–660. <https://doi.org/10.1007/s10548-016-0507-1>, PubMed: 27438589
- Armony, J. L., Chochol, C., Fecteau, S., & Belin, P. (2007). Laugh (or cry) and you will be remembered: Influence of emotional expression on memory for vocalizations. *Psychological Science*, 18, 1027–1029. <https://doi.org/10.1111/j.1467-9280.2007.02019.x>, PubMed: 18031406
- Avero, P., & Calvo, M. G. (2006). Affective priming with pictures of emotional scenes: The role of perceptual similarity and category relatedness. *Spanish Journal of Psychology*, 9, 10–18. <https://doi.org/10.1017/S1138741600005928>, PubMed: 16673618
- Banse, R., & Scherer, K. R. (1996). Acoustic profiles in vocal emotion expression. *Journal of Personality and Social Psychology*, 70, 614–636. <https://doi.org/10.1037/0022-3514.70.3.614>, PubMed: 8851745
- Belin, P., Fillion-Bilodeau, S., & Gosselin, F. (2008). The Montreal Affective Voices: A validated set of nonverbal affect bursts for research on auditory affective processing. *Behavior Research Methods*, 40, 531–539. <https://doi.org/10.3758/BRM.40.2.531>, PubMed: 18522064
- Brosch, T., Grandjean, D., Sander, D., & Scherer, K. R. (2009). Cross-modal emotional attention: Emotional voices modulate early stages of visual processing. *Journal of Cognitive Neuroscience*, 21, 1670–1679. <https://doi.org/10.1162/jocn.2009.21110>, PubMed: 18767920
- Brück, C., Kreifelts, B., & Wildgruber, D. (2011). Emotional voices in context: A neurobiological model of multimodal affective information processing. *Physics of Life Reviews*, 8, 383–403. <https://doi.org/10.1016/j.plrev.2011.10.002>, PubMed: 22035772
- Bublitzky, F., Gerdess, A. B. M., White, A. J., Riemer, M., & Alpers, G. W. (2014). Social and emotional relevance in face processing: Happy faces of future interaction partners enhance the late positive potential. *Frontiers in Human Neuroscience*, 8, 493. <https://doi.org/10.3389/fnhum.2014.00493>, PubMed: 25076881
- Bush, K. A., Privratsky, A., Gardner, J., Zielinski, M. J., & Kilts, C. D. (2018). Common functional brain states encode both perceived emotion and the psychophysiological response to affective stimuli. *Scientific Reports*, 8, 15444. <https://doi.org/10.1038/s41598-018-33621-6>, PubMed: 30337576
- Calvo, M. G., & Nummenmaa, L. (2016). Perceptual and affective mechanisms in facial expression recognition: An integrative review. *Cognition and Emotion*, 30, 1081–1106. <https://doi.org/10.1080/02699931.2015.1049124>, PubMed: 26212348
- Carroll, N. C., & Young, A. W. (2005). Priming of emotion recognition. *Quarterly Journal of Experimental Psychology Section A*, 58, 1173–1197. <https://doi.org/10.1080/02724980443000539>, PubMed: 16194954
- Chang, C.-C., & Lin, C.-J. (2011). LIBSVM: A library for support vector machines. *ACM Transactions on Intelligent Systems and Technology (TIST)*, 2, 1–27. <https://doi.org/10.1145/1961189.1961199>
- Chen, Y.-C., & Spence, C. (2018). Dissociating the time courses of the cross-modal semantic priming effects elicited by naturalistic sounds and spoken words. *Psychonomic Bulletin & Review*, 25, 1138–1146. <https://doi.org/10.3758/s13423-017-1324-6>, PubMed: 28600716
- Chun, J.-W., Park, H.-J., Park, I.-H., & Kim, J.-J. (2012). Common and differential brain responses in men and women to nonverbal emotional vocalizations by the same and opposite sex. *Neuroscience Letters*, 515, 157–161. <https://doi.org/10.1016/j.neulet.2012.03.038>, PubMed: 22465324
- Cichy, R. M., & Pantazis, D. (2017). Multivariate pattern analysis of MEG and EEG: A comparison of representational structure in time and space. *Neuroimage*, 158, 441–454. <https://doi.org/10.1016/j.neuroimage.2017.07.023>, PubMed: 28716718
- Deffke, I., Sander, T., Heidenreich, J., Sommer, W., Curio, G., Trahms, L., et al. (2007). MEG/EEG sources of the 170-ms

- response to faces are co-localized in the fusiform gyrus. *Neuroimage*, 35, 1495–1501. <https://doi.org/10.1016/j.neuroimage.2007.01.034>, PubMed: 17363282
- Delorme, A., & Makeig, S. (2004). EEGLAB: An open source toolbox for analysis of single-trial EEG dynamics including independent component analysis. *Journal of Neuroscience Methods*, 134, 9–21. <https://doi.org/10.1016/j.jneumeth.2003.10.009>, PubMed: 15102499
- Demaree, H. A., Everhart, D. E., Youngstrom, E. A., & Harrison, D. W. (2005). Brain lateralization of emotional processing: Historical roots and a future incorporating “dominance”. *Behavioral and Cognitive Neuroscience Reviews*, 4, 3–20. <https://doi.org/10.1177/1534582305276837>, PubMed: 15886400
- Diéguez-Risco, T., Aguado, L., Albert, J., & Hinojosa, J. A. (2015). Judging emotional congruency: Explicit attention to situational context modulates processing of facial expressions of emotion. *Biological Psychology*, 112, 27–38. <https://doi.org/10.1016/j.biopsycho.2015.09.012>, PubMed: 26450006
- Dobs, K., Isik, L., Pantazis, D., & Kanwisher, N. (2019). How face perception unfolds over time. *Nature Communications*, 10, 1258. <https://doi.org/10.1038/s41467-019-09239-1>, PubMed: 30890707
- Eimer, M., & Holmes, A. (2002). An ERP study on the time course of emotional face processing. *NeuroReport*, 13, 427–431. <https://doi.org/10.1097/00001756-200203250-00013>, PubMed: 11930154
- Filippi, P., Congdon, J. V., Hoang, J., Bowling, D. L., Reber, S. A., Pašukonis, A., et al. (2017). Humans recognize emotional arousal in vocalizations across all classes of terrestrial vertebrates: Evidence for acoustic universals. *Philosophical Transactions of the Royal Society of London, Series B: Biological Sciences*, 284, 20170990. <https://doi.org/10.1098/rspb.2017.0990>, PubMed: 28747478
- Fugate, J. M. B., Gendron, M., Nakashima, S. F., & Barrett, L. F. (2018). Emotion words: Adding face value. *Emotion*, 18, 693–706. <https://doi.org/10.1037/emo0000330>, PubMed: 28604040
- Gao, C., Conte, S., Richards, J. E., Xie, W., & Hanayik, T. (2019). The neural sources of N170: Understanding timing of activation in face-selective areas. *Psychophysiology*, 56, e13336. <https://doi.org/10.1111/psyp.13336>, PubMed: 30710345
- George, M. S., Parekh, P. I., Rosinsky, N., Ketter, T. A., Kimbrell, T. A., Heilman, K. M., et al. (1996). Understanding emotional prosody activates right hemisphere regions. *Archives of Neurology*, 53, 665–670. <https://doi.org/10.1001/archneur.1996.00550070103017>, PubMed: 8929174
- Gerdes, A. B. M., Wieser, M. J., Bublatzky, F., Kusay, A., Plichta, M. M., & Alpers, G. W. (2013). Emotional sounds modulate early neural processing of emotional pictures. *Frontiers in Psychology*, 4, 741. <https://doi.org/10.3389/fpsyg.2013.00741>, PubMed: 24151476
- Giordano, B. L., McAdams, S., Zatorre, R. J., Kriegeskorte, N., & Belin, P. (2013). Abstract encoding of auditory objects in cortical activity patterns. *Cerebral Cortex*, 23, 2025–2037. <https://doi.org/10.1093/cercor/bhs162>, PubMed: 22802575
- Glaser, W. R., & Glaser, M. O. (1989). Context effects in Stroop-like word and picture processing. *Journal of Experimental Psychology: General*, 118, 13–42. <https://doi.org/10.1037/0096-3445.118.1.13>, PubMed: 2522504
- Gong, X., Huang, Y.-X., Wang, Y., & Luo, Y.-J. (2011). Revision of the Chinese Facial Affective Picture System. *Chinese Mental Health Journal*, 25, 40–46.
- Grootswagers, T., Kennedy, B. L., Most, S. B., & Carlson, T. A. (2020). Neural signatures of dynamic emotion constructs in the human brain. *Neuropsychologia*, 145, 106535. <https://doi.org/10.1016/j.neuropsychologia.2017.10.016>, PubMed: 29037506
- Gupta, R. S., Kujawa, A., & Vago, D. R. (2019). The neural chronometry of threat-related attentional bias: Event-related potential (ERP) evidence for early and late stages of selective attentional processing. *International Journal of Psychophysiology*, 146, 20–42. <https://doi.org/10.1016/j.ijpsycho.2019.08.006>, PubMed: 31605728
- Hietanen, J. K., & Astikainen, P. (2013). N170 response to facial expressions is modulated by the affective congruency between the emotional expression and preceding affective picture. *Biological Psychology*, 92, 114–124. <https://doi.org/10.1016/j.biopsycho.2012.10.005>, PubMed: 23131616
- Hinojosa, J. A., Mercado, F., & Carretié, L. (2015). N170 sensitivity to facial expression: A meta-analysis. *Neuroscience & Biobehavioral Reviews*, 55, 498–509. <https://doi.org/10.1016/j.neubiorev.2015.06.002>, PubMed: 26067902
- Killgore, W. D. S., & Yurgelun-Todd, D. A. (2007). The right-hemisphere and valence hypotheses: could they both be right (and sometimes left)? *Social Cognitive and Affective Neuroscience*, 2, 240–250. <https://doi.org/10.1093/scan/nsm020>, PubMed: 18985144
- King, J.-R., & Dehaene, S. (2014). Characterizing the dynamics of mental representations: The temporal generalization method. *Trends in Cognitive Sciences*, 18, 203–210. <https://doi.org/10.1016/j.tics.2014.01.002>, PubMed: 24593982
- Kotz, S. A., & Paulmann, S. (2007). When emotional prosody and semantics dance cheek to cheek: ERP evidence. *Brain Research*, 1151, 107–118. <https://doi.org/10.1016/j.brainres.2007.03.015>, PubMed: 17445783
- Kriegeskorte, N., Mur, M., & Bandettini, P. A. (2008). Representational similarity analysis—Connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2, 4. <https://doi.org/10.3389/neuro.06.004.2008>, PubMed: 19104670
- Lakens, D., & Caldwell, A. R. (2021). Simulation-based power analysis for factorial analysis of variance designs. *Advances in Methods and Practices in Psychological Science*, 4. <https://doi.org/10.1177/2515245920951503>
- Li, Y., Li, S., Hu, W., Yang, L., & Luo, W. (2024). Spatial representation of multidimensional information in emotional faces revealed by fMRI. *Neuroimage*, 290, 120578. <https://doi.org/10.1016/j.neuroimage.2024.120578>, PubMed: 38499051
- Li, Y., Zhang, M., Liu, S., & Luo, W. (2022). EEG decoding of multidimensional information from emotional faces. *Neuroimage*, 258, 119374. <https://doi.org/10.1016/j.neuroimage.2022.119374>, PubMed: 35700944
- Li, S., Zhu, X., Ding, R., Ren, J., & Luo, W. (2019). The effect of emotional and self-referential contexts on ERP responses towards surprised faces. *Biological Psychology*, 146, 107728. <https://doi.org/10.1016/j.biopsycho.2019.107728>, PubMed: 31306692
- Lieenthal, E., Silbersweig, D. A., & Stern, E. (2016). The language, tone and prosody of emotions: Neural substrates and dynamics of spoken-word emotion perception. *Frontiers in Neuroscience*, 10, 506. <https://doi.org/10.3389/fnins.2016.00506>, PubMed: 27877106
- Lin, H., & Liang, J. (2019). Contextual effects of angry vocal expressions on the encoding and recognition of emotional faces: An event-related potential (ERP) study. *Neuropsychologia*, 132, 107147. <https://doi.org/10.1016/j.neuropsychologia.2019.107147>, PubMed: 31325481
- Liu, T., Pinheiro, A., Zhao, Z., Nestor, P. G., McCarley, R. W., & Niznikiewicz, M. A. (2012). Emotional cues during simultaneous face and voice processing: Electrophysiological insights. *PLoS One*, 7, e31001. <https://doi.org/10.1371/journal.pone.0031001>, PubMed: 22383987
- Luo, W., Feng, W., He, W., Wang, N.-Y., & Luo, Y.-J. (2010). Three stages of facial expression processing: ERP study with rapid serial visual presentation. *Neuroimage*, 49, 1857–1867. <https://doi.org/10.1016/j.neuroimage.2009.09.018>, PubMed: 19770052
- Mauchand, M., & Zhang, S. (2023). Disentangling emotional signals in the brain: An ALE meta-analysis of vocal affect

- perception. *Cognitive, Affective, & Behavioral Neuroscience*, 23, 17–29. <https://doi.org/10.3758/s13415-022-01030-y>, PubMed: 35945478
- McCrackin, S. D., & Itier, R. J. (2018). Is it about me? Time-course of self-relevance and valence effects on the perception of neutral faces with direct and averted gaze. *Biological Psychology*, 135, 47–64. <https://doi.org/10.1016/j.biopsycho.2018.03.003>, PubMed: 29524467
- Muukkonen, I., Ölander, K., Numminen, J., & Salmela, V. R. (2020). Spatio-temporal dynamics of face perception. *Neuroimage*, 209, 116531. <https://doi.org/10.1016/j.neuroimage.2020.116531>, PubMed: 31931156
- Muukkonen, I., & Salmela, V. R. (2022). Representational structure of fMRI/EEG responses to dynamic facial expressions. *Neuroimage*, 263, 119631. <https://doi.org/10.1016/j.neuroimage.2022.119631>, PubMed: 36113736
- Mück, M., Ohmann, K., Dummel, S., Mattes, A., Thesing, U., & Stahl, J. (2020). Face perception and narcissism: Variations of event-related potential components (P1 & N170) with admiration and rivalry. *Cognitive, Affective, & Behavioral Neuroscience*, 20, 1041–1055. <https://doi.org/10.3758/s13415-020-00818-0>, PubMed: 32803683
- Nichols, T. E., & Holmes, A. P. (2002). Nonparametric permutation tests for functional neuroimaging: A primer with examples. *Human Brain Mapping*, 15, 1–25. <https://doi.org/10.1002/hbm.1058>, PubMed: 11747097
- Nook, E. C., Lindquist, K. A., & Zaki, J. (2015). A new look at emotion perception: Concepts speed and shape facial emotion recognition. *Emotion*, 15, 569–578. <https://doi.org/10.1037/a0039166>, PubMed: 25938612
- Oosterhof, N. N., Connolly, A. C., & Haxby, J. V. (2016). CoSMoMPPA: Multi-modal multivariate pattern analysis of neuroimaging data in MATLAB/GNU Octave. *Frontiers in Neuroinformatics*, 10, 27. <https://doi.org/10.3389/fninf.2016.00027>, PubMed: 27499741
- Pell, M. D. (2005). Prosody–face interactions in emotional processing as revealed by the facial affect decision task. *Journal of Nonverbal Behavior*, 29, 193–215. <https://doi.org/10.1007/s10919-005-7720-z>
- Pell, M. D., Jaywant, A., Monetta, L., & Kotz, S. A. (2011). Emotional speech processing: Disentangling the effects of prosody and semantic cues. *Cognition & Emotion*, 25, 834–853. <https://doi.org/10.1080/02699931.2010.516915>, PubMed: 21824024
- Pell, M. D., Rothermich, K., Liu, P., Paulmann, S., Sethi, S., & Rigoulot, S. (2015). Preferential decoding of emotion from human non-linguistic vocalizations versus speech prosody. *Biological Psychology*, 111, 14–25. <https://doi.org/10.1016/j.biopsycho.2015.08.008>, PubMed: 26307467
- Pell, M. D., Sethi, S., Rigoulot, S., Rothermich, K., Liu, P., & Jiang, X. (2022). Emotional voices modulate perception and predictions about an upcoming face. *Cortex*, 149, 148–164. <https://doi.org/10.1016/j.cortex.2021.12.017>, PubMed: 35231722
- Pion-Tonachini, L., Kreutz-Delgado, K., & Makeig, S. (2019). ICLabel: An automated electroencephalographic independent component classifier, dataset, and website. *Neuroimage*, 198, 181–197. <https://doi.org/10.1016/j.neuroimage.2019.05.026>, PubMed: 31103785
- Popal, H., Wang, Y., & Olson, I. R. (2019). A guide to representational similarity analysis for social neuroscience. *Social Cognitive and Affective Neuroscience*, 14, 1243–1253. <https://doi.org/10.1093/scan/nsz099>, PubMed: 31989169
- Righart, R., & de Gelder, B. (2006). Context influences early perceptual analysis of faces—An electrophysiological study. *Cerebral Cortex*, 16, 1249–1257. <https://doi.org/10.1093/cercor/bhj066>, PubMed: 16306325
- Righart, R., & de Gelder, B. (2008). Rapid influence of emotional scenes on encoding of facial expressions: An ERP study. *Social Cognitive and Affective Neuroscience*, 3, 270–278. <https://doi.org/10.1093/scan/nsn021>, PubMed: 19015119
- Ross, E. D. (2021). Differential hemispheric lateralization of emotions and related display behaviors: Emotion-type hypothesis. *Brain Sciences*, 11, 1034. <https://doi.org/10.3390/brainsci11081034>, PubMed: 34439653
- Ross, E. D., Thompson, R. D., & Yenkosky, J. (1997). Lateralization of affective prosody in brain and the callosal integration of hemispheric language functions. *Brain and Language*, 56, 27–54. <https://doi.org/10.1006/brln.1997.1731>, PubMed: 8994697
- Sauter, D. A., & Eimer, M. (2010). Rapid detection of emotion from human vocalizations. *Journal of Cognitive Neuroscience*, 22, 474–481. <https://doi.org/10.1162/jocn.2009.21215>, PubMed: 19302002
- Schindler, S., & Bublatzky, F. (2020). Attention and emotion: An integrative review of emotional face processing as a function of attention. *Cortex*, 130, 362–386. <https://doi.org/10.1016/j.cortex.2020.06.010>, PubMed: 32745728
- Schneider, T. R., Engel, A. K., & Debener, S. (2008). Multisensory identification of natural objects in a two-way crossmodal priming paradigm. *Experimental Psychology*, 55, 121–132. <https://doi.org/10.1027/1618-3169.55.2.121>, PubMed: 18444522
- Schupp, H. T., Flaisch, T., Stockburger, J., & Junghöfer, M. (2006). Emotion and attention: Event-related brain potential studies. *Progress in Brain Research*, 156, 31–51. [https://doi.org/10.1016/S0079-6123\(06\)56002-9](https://doi.org/10.1016/S0079-6123(06)56002-9), PubMed: 17015073
- Schupp, H. T., & Kirmse, U. M. (2021). Case-by-case: Emotional stimulus significance and the modulation of the EPN and LPP. *Psychophysiology*, 58, e13766. <https://doi.org/10.1111/psyp.13766>, PubMed: 33483988
- Spreeckelmeyer, K. N., Kutas, M., Urbach, T. P., Altenmüller, E., & Münte, T. F. (2006). Combined perception of emotion in pictures and musical sounds. *Brain Research*, 1070, 160–170. <https://doi.org/10.1016/j.brainres.2005.11.075>, PubMed: 16403462
- Stolz, C., Endres, D., & Mueller, E. M. (2019). Threat-conditioned contexts modulate the late positive potential to faces—A mobile EEG/virtual reality study. *Psychophysiology*, 56, e13308. <https://doi.org/10.1111/psyp.13308>, PubMed: 30548599
- Thom, N., Knight, J., Dishman, R., Sabatinelli, D., Johnson, D. C., & Clementz, B. (2014). Emotional scenes elicit more pronounced self-reported emotional experience and greater EPN and LPP modulation when compared to emotional faces. *Cognitive, Affective, & Behavioral Neuroscience*, 14, 849–860. <https://doi.org/10.3758/s13415-013-0225-z>, PubMed: 24374599
- Tomasello, R., Grisoni, L., Boux, I., Sammler, D., & Pulvermüller, F. (2022). Instantaneous neural processing of communicative functions conveyed by speech prosody. *Cerebral Cortex*, 32, 4885–4901. <https://doi.org/10.1093/cercor/bhab522>, PubMed: 35136980
- Wang, Y., Tang, Z., Zhang, X., & Yang, L. (2022). Auditory and cross-modal attentional bias toward positive natural sounds: Behavioral and ERP evidence. *Frontiers in Human Neuroscience*, 16, 949655. <https://doi.org/10.3389/fnhum.2022.949655>, PubMed: 35967006
- Yuen, K., Johnston, S., Martino, F., Sorger, B., Formisano, E., Linden, D., et al. (2012). Pattern classification predicts individuals' responses to affective stimuli. *Translational Neuroscience*, 3, 278–287. <https://doi.org/10.2478/s13380-012-0029-6>
- Żochowska, A., & Nowicka, A. (2024). Subjectively salient faces differ from emotional faces: ERP evidence. *Scientific Reports*, 14, 3634. <https://doi.org/10.1038/s41598-024-54215-5>, PubMed: 38351111